

# ECS 152A Project 1 – Part 2: Proxy Server

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## Introduction:

This report documents the implementation of a TCP proxy server for Project 1 part 2 of ECS 152A. The proxy server acts as an intermediary between a client and a server, its purpose is to forward messages, and to implement an IP blocklist.

## Initial Attempt:

The initial implementation is pretty similar to the final implementation, except it lacked a timeout system for the proxy server/ server. For the final implementation, the timeout system allows the client to send messages with a 15 second window, if the proxy server/ server does not receive a new message within that window, it will timeout.

## Explanation:

The program consists of three components:

1. Client: Sends JSON message to the proxy server.
2. Proxy Server: Receives the message, checks if IP is in blocklist, if not forwards the message to the server,
3. Server: Receives the message, sends a response back to the proxy server.

## More detail:

The client creates a TCP socket and connects to the proxy server. It sends a JSON message containing the server's IP, port, and message. The client waits 15 seconds for a response before timing out.

The proxy server creates a TCP socket and listens for incoming connections. It receives and parses the JSON message, extracts the server's IP, and checks if it's in the blocklist, if not, it forwards the message to the server. The proxy server waits for the server's response to forward it to the client.

The server creates a TCP socket and listens for incoming connections. It receives the message, creates a response message and sends it to the proxy server.

## Issues Faced:

One issue faced was the initial implementation didn't shut down, so in order to send a new message the IDE had to be restarted. To fix this, the timeout system was added.

## Testing:

### Steps

1. Run proxy\_server.py
2. Run server.py
3. Run client.py

Do this in three different ways commend lines.